

An Integrated Perception, Predictive Beam Control, and Adaptive Optics Platform for UAV Free-Space Optical Communications

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Abstract—Narrow-beam free-space optical (FSO) links mounted on unmanned aerial vehicles (UAVs) require more than an accurate pointing algorithm: they require a sensing, computation, actuation, propagation, and detection chain whose bandwidths and latencies are jointly compatible. This paper presents the hardware realization of a previously introduced software-defined optics framework as an integrated UAV optical-communication platform. A compact perception module combines a BNO08X inertial measurement unit, an OV9281 global-shutter camera, a TF02-Pro LiDAR, and infrared illumination. Multi-rate measurements are fused on embedded computing hardware to obtain continuous position and attitude estimates, while a latency-aware predictor supplies future pose states to a coarse-fine optical controller. Fine correction is performed by a custom dual-axis magnetomotive mirror implemented as a mezzanine stack of four-layer printed circuit boards; transmitter divergence is adjusted independently through a piezoelectric optical module. The optical channel is represented in Zemax OpticStudio using component-level transmitter and receiver models, UAV motion, atmospheric attenuation, and Kolmogorov turbulence. At the receiver, a telecom-band InGaAs detector evaluates link power, signal-to-noise ratio, and bit-error-rate trends, while a filtered dual-silicon-photodiode unit provides an auxiliary alignment channel under ambient illumination. Bench measurements demonstrate sustainable sensing rates of 400 Hz (IMU), 100 Hz (LiDAR), and 30 Hz (vision), a 35.61 ms fine-steering response, a 145.78–170.47 ms coarse-servo response, approximately $2.5\times$ lower pose error than IMU-only dead reckoning, and clear separation of the auxiliary laser signal from the ambient-light noise ceiling. The result is a hardware–software co-designed platform that connects predictive optical control to physical UAV-ready sensing and actuation.

Index Terms—Free-space optical communication, unmanned aerial vehicle, sensor fusion, predictive control, beam steering, MEMS mirror, adaptive divergence, photodetection.

I. INTRODUCTION

FREE-SPACE optical communication offers high capacity, license-free spectrum, electromagnetic immunity, and strong spatial confinement, making it attractive for aerial networking and inter-UAV links [1], [2]. These advantages come with a severe alignment constraint. Translational vibration, attitude jitter, gimbal lag, structural resonance, and atmospheric turbulence can move or broaden the beam footprint faster than a conventional camera-feedback loop can correct it. Because an aperture only collects the portion of the irradiance distribution that remains inside its boundary, even a short interruption in pointing can cause a disproportionate loss in received power.

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Pointing, acquisition, and tracking (PAT) research has therefore investigated gimbals, inertial aiding, quadrant detectors, and learning-assisted control [3]–[6]. Most reported architectures, however, isolate one layer of the problem: an algorithm is evaluated without the actuator dynamics, a steering mechanism is tested without multi-modal motion estimation, or the optical channel is simulated without a practical receiver. Our earlier software-defined optics (SDO) study addressed the algorithmic gap by using a Double Deep Q-Network (DDQN) to predict UAV motion and issue feedforward steering commands before the corresponding misalignment reached the receiver [7]. That work established predictive-control feasibility and evaluated coupling through Zemax OpticStudio, but much of the sensing, steering, beam-shaping, and receiver chain remained simulated or represented by commercial-equivalent blocks.

This paper advances that work from an algorithm-centered demonstration to a hardware–software co-designed platform. The system is organized as a continuous chain:

$$\begin{aligned} \text{perception} &\rightarrow \text{fusion} \rightarrow \text{prediction} \\ &\rightarrow \text{pointing/divergence} \rightarrow \text{channel} \rightarrow \text{detection.} \end{aligned} \quad (1)$$

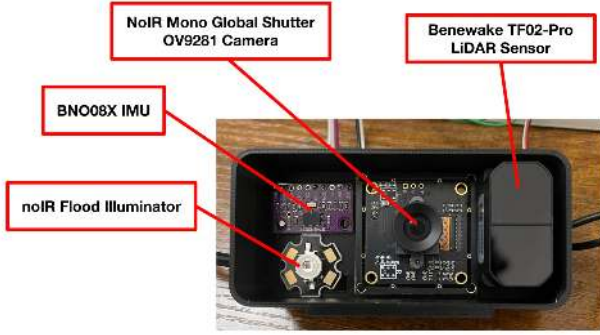
The main contributions are as follows.

- (i) A compact visual–inertial–LiDAR module and a multi-rate embedded architecture are integrated to provide high-rate motion observations and bounded low-frequency drift.
- (ii) A latency-aware interface connects future pose estimates to coarse gimbal motion and a custom fine-steering mirror, allowing sensing, prediction horizon, and actuator response to be considered in the same timing budget.
- (iii) A dual-axis magnetomotive steering mirror is realized as a three-board mezzanine stack of four-layer PCBs and is characterized using camera-based full-field calibration, circular-scan validation, and sensor-to-optical-response timing.
- (iv) Pointing control is paired with piezoelectric divergence control, component-level optical-channel simulation, a telecom-band detector model, and a separate filtered silicon-photodiode alignment receiver.

Table I summarizes the distinction between the prior SDO study and the present hardware realization. The objective is not to replace the predictive-control results, but to expose and validate the physical interfaces that determine whether those results can be realized on a UAV payload.

TABLE I
EVOLUTION FROM THE PREDICTIVE SDO STUDY TO THE INTEGRATED HARDWARE PLATFORM

Layer	Earlier predictive SDO study	Hardware realization in this work
Motion sensing	IMU-driven motion sequence used for forecasting	BNO08X IMU, global-shutter vision, TF02-Pro LiDAR, and IR illumination in one compact perception module
State estimation	Motion variables supplied to the predictor	Multi-rate embedded fusion of inertial, visual, and absolute-range observations
Computation	Raspberry Pi inference with periodic external training	Raspberry Pi supervisory compute plus STM32/ESP32 deterministic sensor and actuator interfaces
Beam control	Predicted correction applied to a MEMS-equivalent steering block	Coarse servo pointing, calibrated custom PCB magnetomotive fine steering, and adaptive piezoelectric divergence
Optical channel	Telemetry-driven Zemax ray tracing and receiver-aperture coupling	Component-level transmitter/receiver model with loss, turbulence, detector SNR, and BER evaluation
Receiver feedback	Primarily simulated receiver-plane power	Telecom-band InGaAs power path plus an auxiliary filtered dual-silicon-photodiode alignment channel



(a) Annotated sensor placement in the compact perception enclosure.



(b) CAD-to-hardware comparison of the enclosure and assembled module.

Fig. 1. Visual-inertial-LiDAR perception module. The BNO08X, OV9281 camera, TF02-Pro LiDAR, and IR flood illumination share a compact rigid housing.

II. INTEGRATED HARDWARE ARCHITECTURE

A. Perception Module

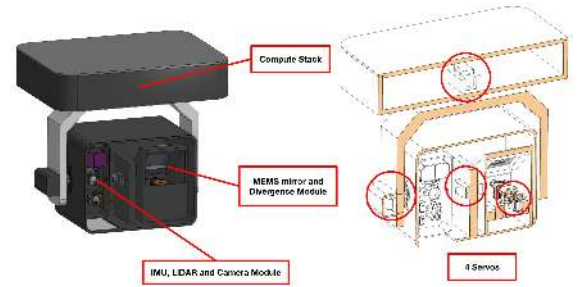
The perception head packages four complementary sensing elements inside a compact enclosure: a BNO08X inertial measurement unit (IMU), a monochrome OV9281 global-shutter camera, a Benewake TF02-Pro single-point LiDAR, and a 940-nm infrared flood illuminator. The IMU provides high-rate acceleration, angular velocity, and orientation products; the camera supplies horizontal image motion and scene features without rolling-shutter skew; and the LiDAR supplies an absolute range reference along the module axis. The infrared source extends feature visibility under dim conditions without sharing the telecom data wavelength.

The layout in Fig. 1 places the camera and LiDAR close to one another to reduce parallax, while the IMU is mounted on the same rigid body so that optical and inertial measurements experience nearly the same motion. This shared mechanical reference is important: an algorithm cannot remove a relative motion artifact introduced by flexure between sensors unless that flexure is independently modeled.

B. Embedded Compute, Power, and Communication

The embedded stack separates high-level estimation from deterministic input/output. A Raspberry Pi executes vision, data logging, supervisory fusion, and predictor inference. An STM32F446RE or ESP32 handles timestamped sensor acquisition, pulse-width-modulated actuator commands, and low-level control. The STM32F446RE provides an Arm Cortex-M4 processor operating at up to 180 MHz [8]; in the current bring-up, the BNO085 uses 400-kHz I²C with data-ready interrupts, while actuator commands are produced by hardware timers. This partition avoids allowing camera processing or file I/O on the single-board computer to perturb the actuator schedule.

Figure 2(a) shows the power and communication topology. The optical path, steering mirror, divergence stage, servos, perception module, and processor share a regulated 5-V distribution network with local conversion where required. Figure 2(b) illustrates the packaged compute and lower sensor-control payload. The power drawing is an engineering integration diagram rather than a final flight-qualified schematic; connector retention, electromagnetic compatibility, fusing, and



Let $\Delta \mathbf{p}_\perp = [\Delta y, \Delta z]^\top$ be the predicted displacement normal to a link of length L . For small angles, the required

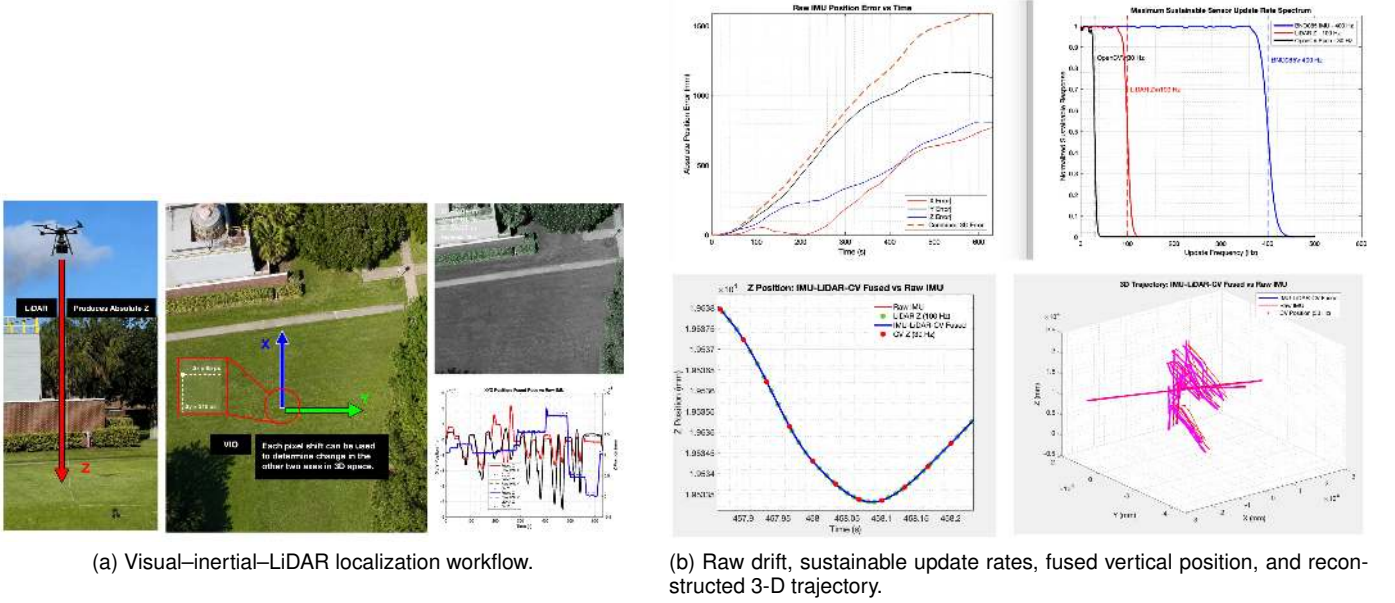


Fig. 3. Multi-rate pose estimation. The IMU supplies high-rate propagation, LiDAR provides an absolute ranging constraint, and visual features bound horizontal drift.

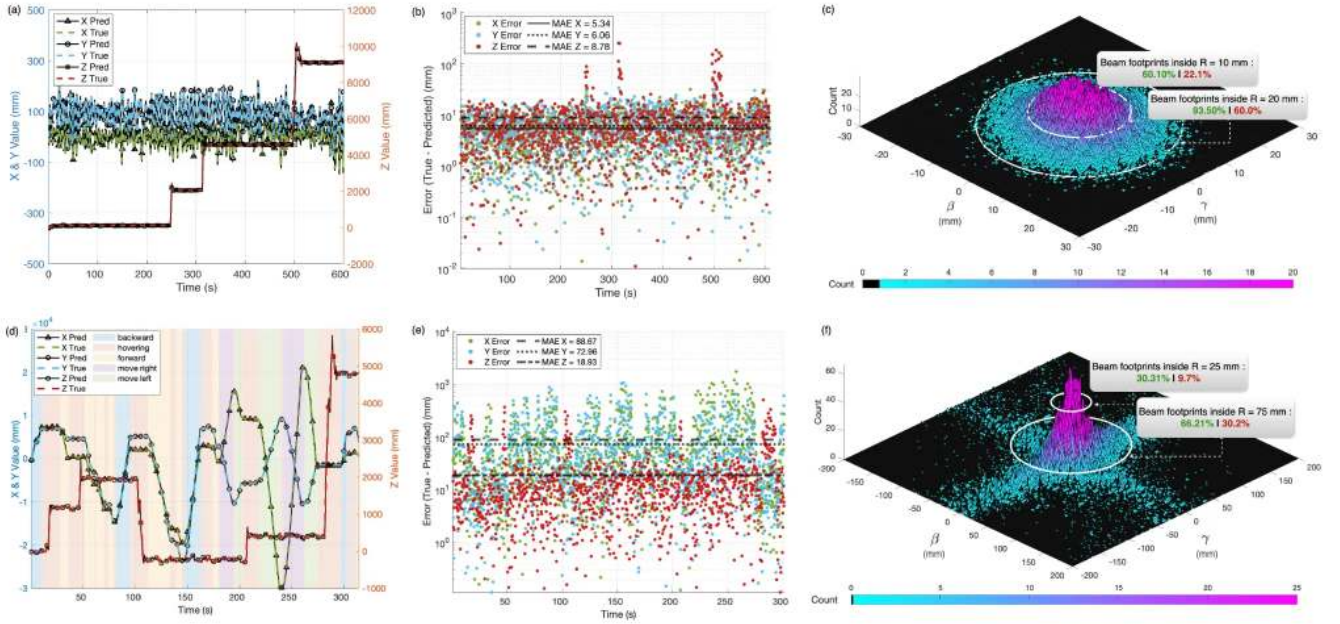


Fig. 4. Prior DDQN predictive-control validation used as the input benchmark for the hardware realization: trajectory estimates, prediction errors, and corrected beam-footprint distributions during hovering (top) and translation (bottom) [7].

line-of-sight correction is

$$\theta_{\text{LOS}} \approx \frac{\Delta p_{\perp}}{L}. \quad (6)$$

The command is separated into a low-frequency component for the gimbal and a high-frequency residual for the fine mirror. Because reflection doubles the optical deflection, the mirror mechanical angle is approximately half the required residual beam angle. The measured servo latency of 145.78–170.47 ms corresponds to only 5.87–6.86 effective updates per second, whereas the fine-steering path responds in 35.61 ms (about 28.1 Hz). These measured dynamics justify the two-

stage controller rather than treating the gimbal and mirror as interchangeable actuators.

B. Magnetomotive PCB Steering Mirror

The fine-steering mechanism is a custom dual-axis magnetomotive mirror. Four planar coils are implemented on each four-layer PCB, and three boards are stacked as a mezzanine assembly. A compliant center structure carries a first-surface mirror and small neodymium magnets. Differential coil excitation produces torque about two axes, while the layered stack

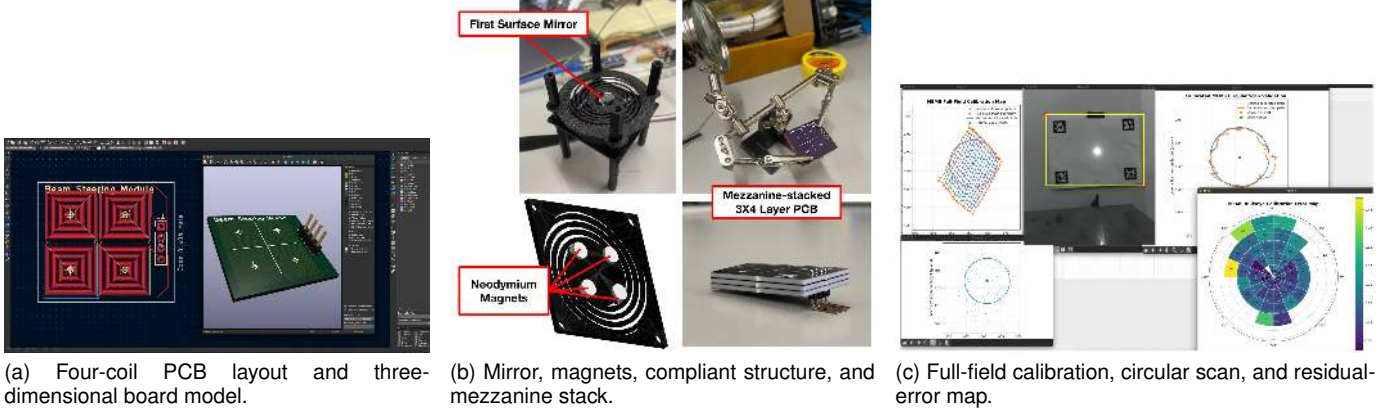


Fig. 5. Custom magnetomotive fine-steering mirror and camera-based characterization. The calibrated operating region is used to convert predicted optical displacement into bounded PWM commands.

increases usable magnetic interaction without the cost and packaging complexity of a commercial fast-steering mirror.

Camera calibration maps the two PWM drive coordinates (u_x, u_y) to the observed laser location (c_x, c_y) . The controller uses the inverse of a fitted calibration surface,

$$[u_x, u_y]^T = f_{\text{cal}}^{-1}(c_x^*, c_y^*), \quad (7)$$

and constrains commands to the largest calibrated circle. Figure 5 shows the PCB layout, physical stack, full-field sampling, commanded circular scan, and residual error map. In timing tests using a 120.64-frame/s global-shutter camera, the command-to-visible response was 31.29 ms and the command-to-settled response was 38.63 ms; their representative mean is 35.61 ms.

C. Piezoelectric Divergence Control

Pointing determines where the beam center lands; divergence determines how the available optical power is distributed around that center. For a Gaussian launch beam with waist w_0 , the nominal radius after distance L is

$$w(L) = w_0 \sqrt{1 + \left(\frac{\lambda L}{\pi w_0^2} \right)^2}. \quad (8)$$

A piezoelectric optical module changes the launch condition and therefore w_0 . The control objective is to select a divergence that is narrow enough to maximize aperture capture but broad enough to retain useful power under the predicted residual pointing distribution. For an aligned circular receiver of radius a , a first-order capture approximation is

$$\eta_{\text{ap}} \approx 1 - \exp \left[-\frac{2a^2}{w_{\text{eff}}^2(L)} \right], \quad (9)$$

where w_{eff} includes diffraction and turbulence-induced spreading. The experimental module demonstrated up to a $2.5\times$ increase in received-power efficiency relative to a fixed launch condition over the evaluated link geometries. In the integrated controller, divergence is updated on the slow range scale, while the mirror handles high-rate centroid displacement.

V. OPTICAL CHANNEL AND RECEIVER

A. Zemax Link Model

The optical model traces a 1310-nm transmitter through a fiber collimator, a $5\times$ Galilean beam expander, the steering element, the free-space path, and a Newtonian receiver to the detector plane. Commercial component prescriptions are used where available [9]–[11]. UAV motion is inserted through coordinate breaks controlled through the Zemax OpticStudio API. Atmospheric loss is represented by a Beer–Lambert transmission term $T_{\text{atm}} = \exp(-\alpha L)$, and random Kolmogorov phase screens reproduce turbulence-induced spatial intensity modulation.

If $I(y, z, t)$ is the detector-plane irradiance, the power captured by aperture $\mathcal{A}$ is

$$P_{\text{ap}}(t) = \iint_{\mathcal{A}} I(y, z, t) dy dz, \quad (10)$$

and the power delivered to the communications detector is

$$P_d(t) = P_{\text{ap}}(t) \eta_{\text{rx}} 10^{-A/10}, \quad (11)$$

where η_{rx} is the receiver-optics transmission and $A = 20$ dB is the attenuation stage used in the detector-level analysis.

B. Communications and Alignment Detection

The communications analysis uses a Thorlabs PDA20C2 InGaAs amplified detector, which covers 800–1700 nm and has a 5-MHz bandwidth and 22 pW/ $\sqrt{\text{Hz}}$ noise-equivalent power [12]. At 1550 nm, the detector responsivity is approximately 1 A/W and the 50- Ω transimpedance gain is 175 kV/A. With responsivity $\mathcal{R}$ and electrical bandwidth B , the detector-current SNR is estimated as

$$\text{SNR}(t) = \frac{[\mathcal{R} P_d(t)]^2}{(\text{NEP } \mathcal{R})^2 B}, \quad (12)$$

with receiver-electronics noise added when measured. For a Gaussian decision statistic, the corresponding uncoded binary-error estimate is

$$\text{BER}(t) = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{\text{SNR}(t)}{2}} \right). \quad (13)$$

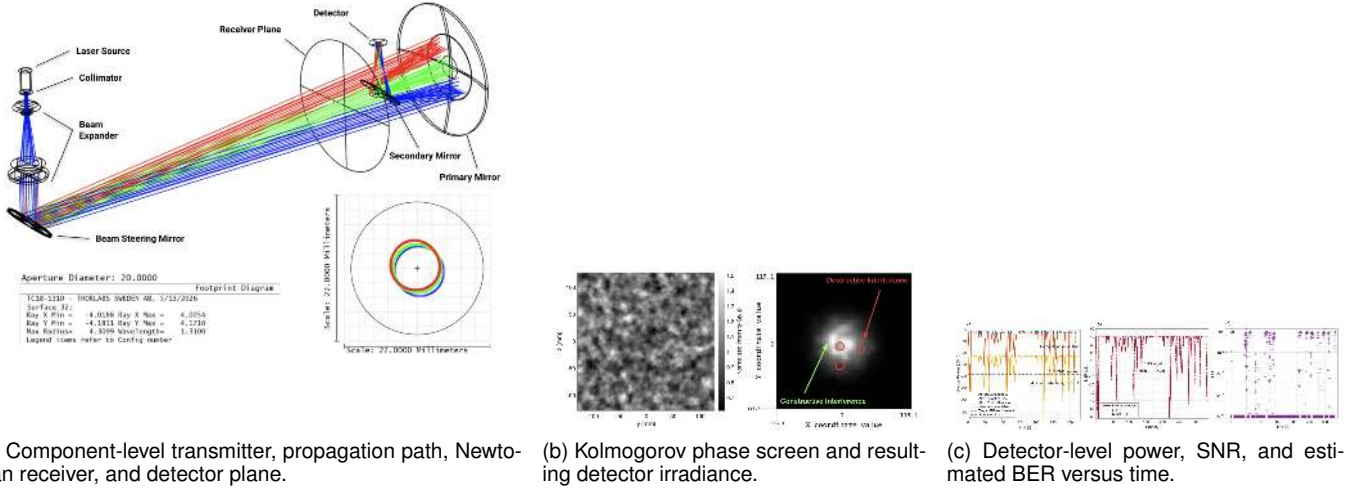


Fig. 6. Optical channel and detector-level evaluation. UAV motion, atmospheric attenuation, and turbulence are propagated to the receiver plane before the collected power is passed through receiver loss and attenuation.

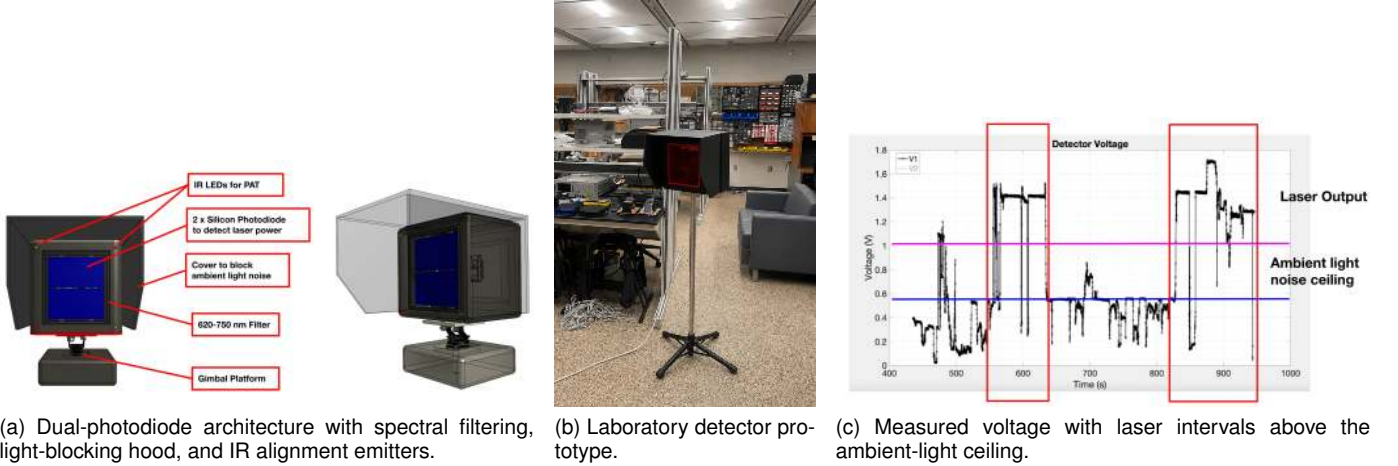


Fig. 7. Auxiliary optical alignment receiver. The filtered silicon detector is spectrally and functionally separated from the telecom-band communications detector.

These expressions translate the optical footprint into detector-domain margins and expose interruptions that aperture coupling alone can hide.

The separate dual-silicon-photodiode receiver in Fig. 7 is an auxiliary PAT and power-monitoring channel. Its 620–750-nm filter, deep light-blocking hood, and integrated IR illumination are not intended to replace the 1310/1550-nm InGaAs communications detector. Instead, the module provides an inexpensive spatially filtered alignment observable. In laboratory measurements, laser-on intervals rise above an approximately 0.55-V ambient-light ceiling and frequently exceed a 1.0-V decision threshold, demonstrating that the optical event can be separated from background variations.

VI. SYSTEM-LEVEL EVALUATION AND DISCUSSION

Table III collects the quantitative results available across the integrated prototype. Together, they show that each layer satisfies a different part of the control problem. The sensor rates provide sufficient temporal density for a 50-ms

future-state estimate. The fine mirror can respond inside that horizon, while the slower gimbal cannot and must receive a lower-frequency command. Sensor fusion limits the drift that would otherwise bias the predictor, and the auxiliary detector produces a direct optical observable for acquisition and alignment. Finally, the Zemax model converts residual pointing and divergence into received-power, SNR, and BER trends.

The prototype also identifies integration limits that are less visible in simulation. First, all measurements require a common time base; otherwise, sensor fusion may improve nominal trajectory smoothness while degrading prediction alignment. Second, camera-derived displacement depends on scene texture, altitude, illumination, and calibration. The IR illuminator improves local visibility but cannot guarantee outdoor feature quality over the full link range. Third, the PCB mirror’s pixel-domain calibration must be converted to angular response and periodically checked for temperature, mounting, and hysteresis drift. Fourth, the reported divergence

TABLE III
MEASURED AND PREVIOUSLY VALIDATED PERFORMANCE OF THE INTEGRATED PLATFORM

Subsystem	Result	System implication
Multi-rate sensing	400/100/30 Hz IMU/LiDAR/vision	High-rate propagation with periodic absolute and visual corrections
Pose fusion	$\sim 2.5\times$ lower error than IMU-only	Reduces long-term pointing bias presented to the predictor
Fine steering	31.29 ms visible; 38.63 ms settled; 35.61 ms representative	Compatible with a 50-ms fine-correction prediction horizon
Coarse steering	145.78–170.47 ms	Requires lower-bandwidth or longer-horizon commands
Prior predictive control	93.5% within 20 mm (hover); 66.2% within 75 mm (motion)	Establishes the feedforward beam-confinement benefit [7]
Adaptive divergence	Up to $2.5\times$ received-power efficiency	Adds a power-capture degree of freedom independent of centroid steering
Auxiliary detector	Laser-on voltage above ambient ceiling	Provides receiver-side acquisition/alignment feedback

gain must be characterized over a documented range/aperture matrix before it can be generalized to flight. Finally, a binary error estimate derived from detector SNR is not a substitute for a modulated end-to-end data test.

Accordingly, the next validation stage will combine synchronized truth instrumentation, the multi-rate estimator, the predictor, both pointing stages, the divergence actuator, and a modulated optical channel in one closed-loop experiment. The principal evaluation variables should be residual centroid error, captured-power outage probability, acquisition time, closed-loop bandwidth, energy consumption, and BER under controlled platform motion and turbulence. Outdoor UAV-to-UAV testing should follow only after optical safety, mechanical retention, and flight-worthiness reviews.

VII. CONCLUSION

This paper presented the hardware realization of a predictive software-defined optics concept for UAV free-space optical communication. The platform connects a compact visual–inertial–LiDAR module to multi-rate pose fusion, future-state prediction, coarse gimbal pointing, a custom PCB-based magnetomotive fine-steering mirror, piezoelectric divergence control, component-level optical propagation, and two complementary receiver functions. Bench results establish 400/100/30-Hz sensing, approximately $2.5\times$ pose-estimation improvement over inertial dead reckoning, a 35.61-ms fine-steering response, and optical event detection above ambient background. The hardware measurements clarify a key design rule: the prediction horizon and actuator bandwidth must be allocated together, with the fine mirror rejecting fast motion, the gimbal following slow motion, and divergence adapting to range and residual uncertainty. The resulting architecture extends predictive beam steering from a simulation-centered framework toward an experimentally grounded, UAV-ready optical communication platform.

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